

Relation Between Fermi–Amaldi and Exact Exchange Energy Densities

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I. MAIN TEXT

Consider the exact exchange energy density (closed-shell):

$$e_x^{\text{exact}}(\mathbf{r}) = -\frac{1}{4} \int \frac{|\gamma(\mathbf{r}; \mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}', \quad (1)$$

where

$$\gamma(\mathbf{r}, \mathbf{r}') = 2 \sum_{i=1}^{N/2} \varphi_i^*(\mathbf{r}) \varphi_i(\mathbf{r}') \quad (2)$$

is the one-electron reduced density matrix [1, 2], and the Fermi–Amaldi [3, 4] energy density expressed via electron density, $\rho(\mathbf{r}) \geq 0$, and electron number, N ,

$$e_x^{\text{FA}}(\mathbf{r}) = -\frac{1}{2N} \int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}'. \quad (3)$$

For one-orbital systems (one-electron and two-electron singlet), the two quantities are equal for all values of $\mathbf{r}$ [5–7]:

$$e_x^{\text{exact}}(\mathbf{r}) = e_x^{\text{FA}}(\mathbf{r}) \quad N \leq 2 \quad \text{for all } \mathbf{r}. \quad (4)$$

For systems with more than one molecular orbital, the spatial behavior of both quantities needs to be studied closer to establish a relationship between the exact and the Fermi–Amaldi exchange energy densities.

For two integrals with different integrands $f(x)$ and $g(x)$, on the interval $[a, b]$

$$\int_a^b f(x) dx \geq \int_a^b g(x) dx, \quad \text{when } f(x) \geq g(x). \quad (5)$$

Therefore, since $1/|\mathbf{r} - \mathbf{r}'| > 0$ everywhere in space and is the same for both exchange energy densities, by the

Cauchy–Schwarz inequality:

$$\rho(\mathbf{r})\rho(\mathbf{r}') \geq |\gamma(\mathbf{r}, \mathbf{r}')|^2 \quad \text{for all } \mathbf{r}, \mathbf{r}', \quad (6)$$

leading to the inequality

$$\int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' \geq \int \frac{|\gamma(\mathbf{r}, \mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' \quad \text{for all } \mathbf{r}. \quad (7)$$

Applying the definitions (1) and (3), we arrive at another inequality:

$$\frac{e_x^{\text{FA}}(\mathbf{r})}{e_x^{\text{exact}}(\mathbf{r})} \geq \frac{2}{N} \quad \text{for all } \mathbf{r}. \quad (8)$$

As $|\mathbf{r}| \rightarrow \infty$, the denominator can be approximated as $|\mathbf{r} - \mathbf{r}'| \approx |\mathbf{r}|$, revealing that both exchange energy densities approach the same asymptotic limit:

$$e_x^{\text{exact}}(\mathbf{r}) \sim e_x^{\text{FA}}(\mathbf{r}) \sim -\frac{\rho(\mathbf{r})}{2|\mathbf{r}|}. \quad (9)$$

Thus, in the asymptotic limit:

$$\lim_{|\mathbf{r}| \rightarrow \infty} \frac{e_x^{\text{FA}}}{e_x^{\text{exact}}} = 1. \quad (10)$$

Conversely, in the limit of infinitesimally small inter-electronic distances ($\mathbf{r} \rightarrow \mathbf{r}'$), the spatial parts of both integrands approach equality:

$$|\gamma(\mathbf{r}, \mathbf{r}')|^2 \rightarrow \rho(\mathbf{r})\rho(\mathbf{r}') \quad \text{as } \mathbf{r}' \rightarrow \mathbf{r}. \quad (11)$$

However, because the exact exchange prefactor, $1/4$, is larger than the Fermi–Amaldi prefactor, $1/(2N)$, for $N > 2$, the region where $|\mathbf{r} - \mathbf{r}'| \rightarrow 0$ is dominated by the exact exchange energy density:

$$\lim_{|\mathbf{r}| \rightarrow |\mathbf{r}'|} \frac{e_x^{\text{FA}}}{e_x^{\text{exact}}} = 0 \quad \text{for large } N. \quad (12)$$

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